



Data Scientist focused on machine learning, probabilistic modelling, and transforming unstructured data into structured representations. Applied experience in climate analytics, predictive maintenance, and operational optimization.

SKILLS

Programming & Tools	Python (pandas, scikit-learn), SQL, R, Java, Git, Docker
Data Engineering	Spark, Hadoop, Airflow, ETL pipelines, web scraping (Playwright)
Machine Learning	Supervised learning, NLP, feature engineering, model evaluation, XGBoost, Random Forest
Optimization & Stats	Statistical modeling, econometrics, constrained optimization
Visualization	Matplotlib, Plotly, Streamlit
Languages	Dutch (Native), English (Fluent), Spanish (Intermediate)

TECHNICAL EXPERIENCE

Research Intern, Climate & Machine Learning Deltares	Jan 2026 — Present Utrecht, Netherlands
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- Built an end-to-end drought impact analysis pipeline from news collection and event detection to predictive modeling.
- Developed an interactive visualization platform for identifying drought-vulnerable regions, supporting decision-making by stakeholders including Rijkswaterstaat.
- Developed an NLP workflow for feature extraction, event classification, and location detection from unstructured news data.
- Trained machine learning models (including Random Forest) to predict drought impact from combined climate and socio-economic features.

Data Scientist (Freelance) Kuijpers BV	June 2025 — Sep 2025 's-Hertogenbosch, Netherlands
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- Developed a proof-of-concept predictive maintenance approach by extracting structured state-space features (e.g., machine state and actions) from unstructured service text.
- Built and evaluated few-shot machine learning models for information extraction on labeled maintenance records.
- Coordinated data labeling with domain experts at Kuijpers BV to construct ground-truth datasets for model training.

Data Scientist (Part-time) Déhora Consultancy Group	March 2023 — May 2023 Amsterdam, Netherlands
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- Developed a constraint-based crew allocation algorithm for Fire Department staffing, ensuring required operational role coverage per team.
- Modeled personnel assignment as a structured optimization problem with role and grouping constraints.

TECHNICAL PROJECTS

Reinforcement Learning: Stochastic Scheduling	TU/e
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- Solved stochastic maintenance scheduling problems using **Value Iteration**, **Policy Iteration**, and **Q-Learning**.

Deep Learning: Grid Architectures	TU/e
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- Designed neural networks based on data grids and symmetries to fit the use case efficiently.

Generative AI: Transformer Implementation	TU/e
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- Built an **Autoregressive Transformer** from scratch; implemented attention blocks and tuned with temperature scaling.

Competitive Sudoku Agent	TU/e
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- Developed a Sudoku solver using **Minimax** and **Markov Chain Monte Carlo (MCMC)** for state exploration.

EDUCATION

MSc Data Science & Artificial Intelligence , Eindhoven University of Technology	Sept 2024 — Present
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- *In Progress: 90/120 ECTS completed. Focus on fundamental AI and mathematical analysis.*
- *Minor: Data Science in Business and Entrepreneurship (Jheronimus Academy of Data Science) — 20 ECTS.*

BSc Data Science (Joint Degree) , TU Eindhoven & Tilburg University	Sept 2020 — June 2023
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- *Multidisciplinary foundation in Econometrics, Computer Science, Statistics and Business.*
- *Minor: Physics / Geoscience (Utrecht University) — 50 ECTS.*

ACTIVITIES

Athletics: Run Club & Gym	2025 — Present
Music: Piano & Salsa	2015 — Present
Strategic Games: Chess Club	2023 — 2025
Student Committees: Sports & First-Year	2020 — 2023